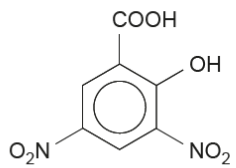
Question				Marking details	Marks available AO1	AO2	AO3	Total	Maths	Prac
		(ii)		mole ratio 1:1 moles of 2-hydroxybenzoic acid used = $\frac{4.00}{138.06}$ = 0.0290 (1) theoretical moles of nitroacid = 0.290 theoretical mass of nitroacid = 5.306 / 5.31 g (1) percentage yield is 41 therefore mass obtained = $\frac{5.31 \times 41}{100}$ = 2.18 g (1)		3		3	2	
		(iii)		relative mass of compound **J** without two X groups ($C_7H_6O_3$) = 136 relative mass of two X groups is 228 – 136 = 92 each X group has mass of 46 (1) award (1) for either of following • X must contain 2 oxygen atoms (as 7 in total) atoms therefore remainder is 14 - other atom must be nitrogen and X is NO_2 • structure of compound **J** 			2	2		
		(iv)		lower temperature / lower concentration HNO_3 / lower volume of aqueous HNO_3 / less heating time			1	1		
	(b)			2-hydroxybenzenecarboxylic acid will react with $NaHCO_3$ / Na_2CO_3 to give effervescence	1			1		1
				Question 11 total	**5**	**3**	**5**	**13**	**2**	**7**